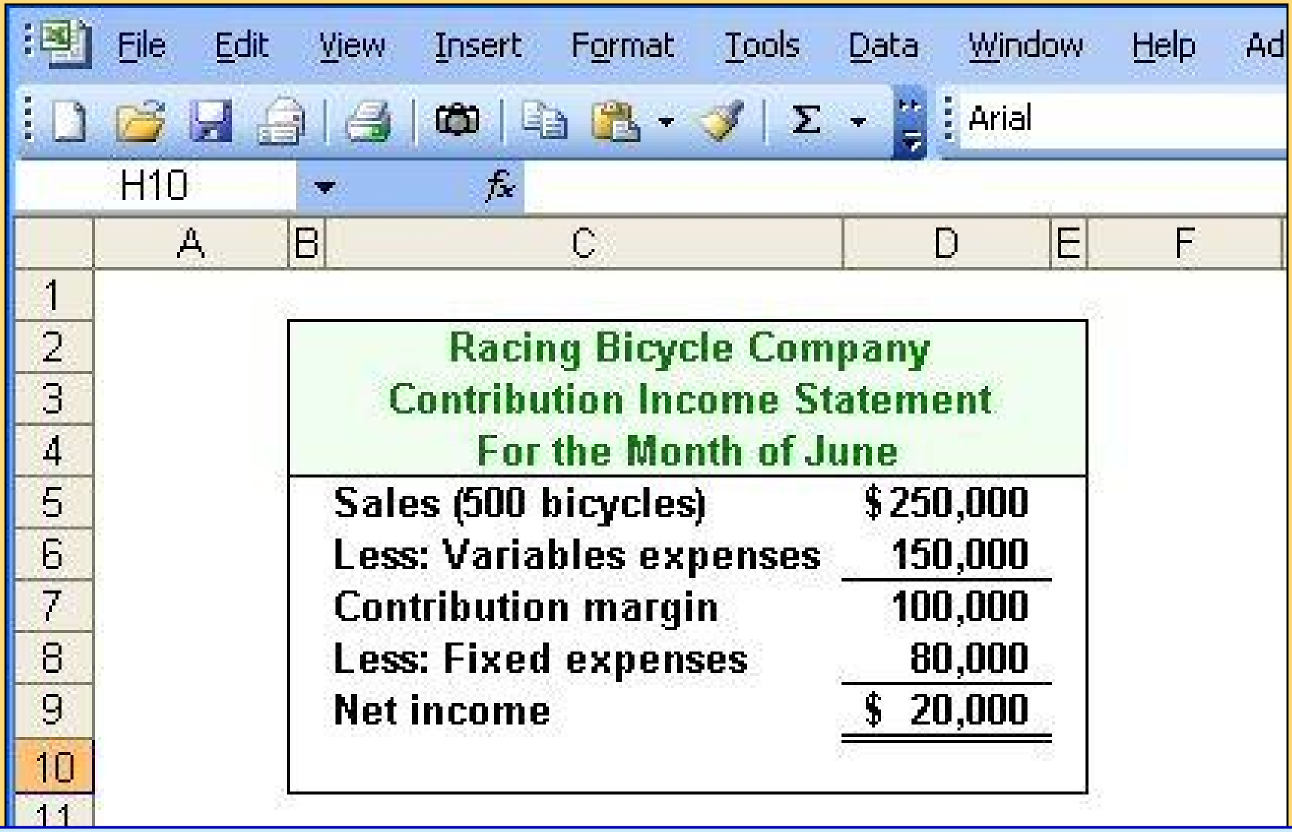
Cost-Volume-Profit Relationships

Chapter Six

Learning Objective 1

Explain how changes in activity affect contribution margin and net operating income.

Basics of Cost-Volume-Profit Analysis



Racing Bicycle Company Contribution Income Statement For the Month of June	
Sales (500 bicycles)	$ 250,000
Less: Variables expenses	150,000
Contribution margin	100,000
Less: Fixed expenses	80,000
Net income	$ 20,000

Contribution Margin (CM) is the amount remaining from sales revenue after variable expenses have been deducted.

Basics of Cost-Volume-Profit Analysis

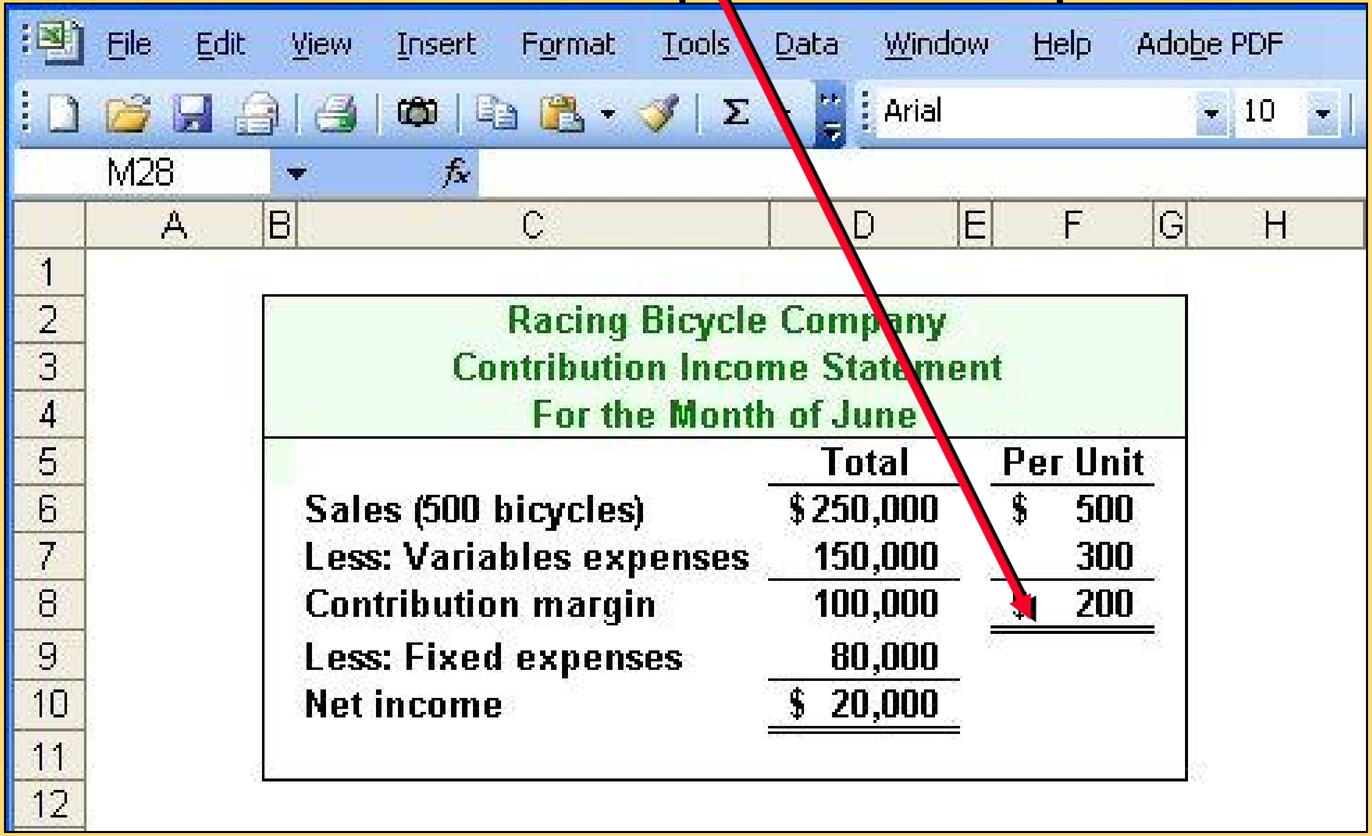
Racing Bicycle Company
Contribution Income Statement
For the Month of June

Sales (500 bicycles)	$ 250,000
Less: Variables expenses	150,000
Contribution margin	100,000
Less: Fixed expenses	80,000
Net income	$ 20,000

CM is used first to cover fixed expenses. Any remaining CM contributes to net operating income.

The Contribution Approach

Sales, variable expenses, and contribution margin can also be expressed on a per unit basis. If Racing sells an additional bicycle, $200 additional CM will be generated to cover fixed expenses and profit.



Racing Bicycle Company
Contribution Income Statement
For the Month of June

	Total	Per Unit
Sales (500 bicycles)	$ 250,000	$ 500
Less: Variables expenses	150,000	300
Contribution margin	100,000	$ 200
Less: Fixed expenses	80,000	
Net income	$ 20,000	

The Contribution Approach

Each month, Racing must generate at least $80,000 in total CM to break even.

Racing Bicycle Company
Contribution Income Statement
For the Month of June

	Total	Per Unit
Sales (500 bicycles)	$ 250,000	$ 500
Less: Variables expenses	150,000	300
Contribution margin	100,000	$ 200
Less: Fixed expenses	80,000	
Net income	$ 20,000	

The Contribution Approach

If Racing sells **400 units** in a month, it will be operating at the *break-even point*.

Racing Bicycle Company
Contribution Income Statement
For the Month of June

	Total	Per Unit
Sales (400 bicycles)	$ 200,000	$ 500
Less: Variables expenses	120,000	300
Contribution margin	80,000	$ 200
Less: Fixed expenses	80,000	
Net income	$ -	

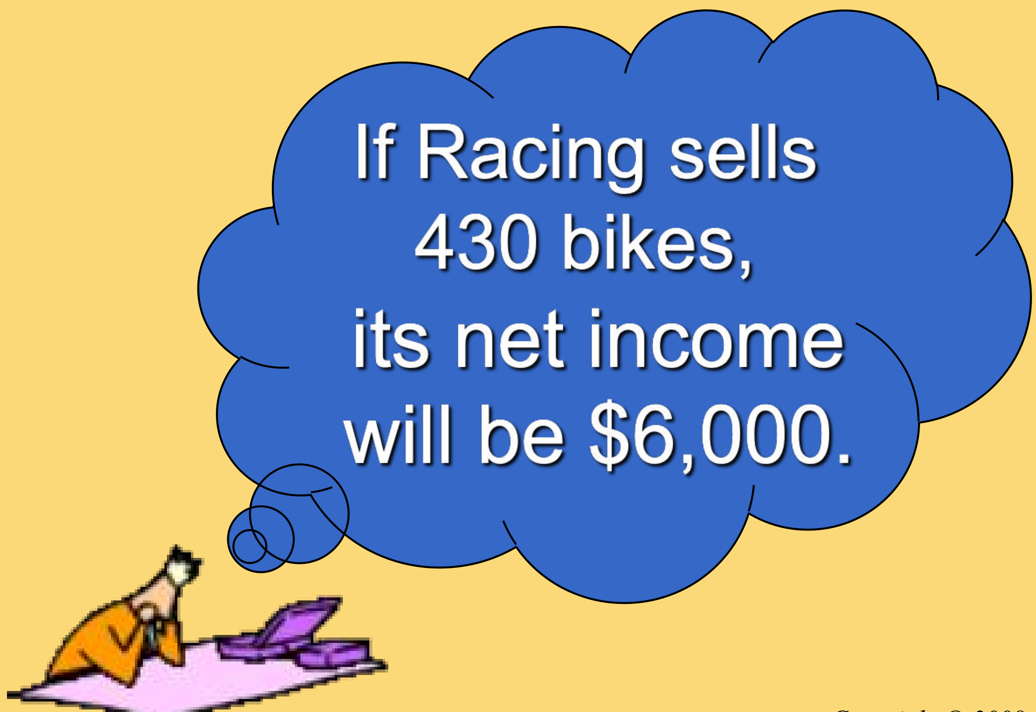
The Contribution Approach

If Racing sells one more bike (401 bikes), net operating income will increase by $200.

Racing Bicycle Company Contribution Income Statement For the Month of June	Total	Per Unit
Sales (401 bicycles)	$ 200,500	$ 500
Less: Variables expenses	120,300	300
Contribution margin	80,200	$ 200
Less: Fixed expenses	80,000	
Net income	$ 200	

The Contribution Approach

We do not need to prepare an income statement to estimate profits at a particular sales volume. Simply multiply the number of units sold above break-even by the contribution margin per unit.



Learning Objective 2

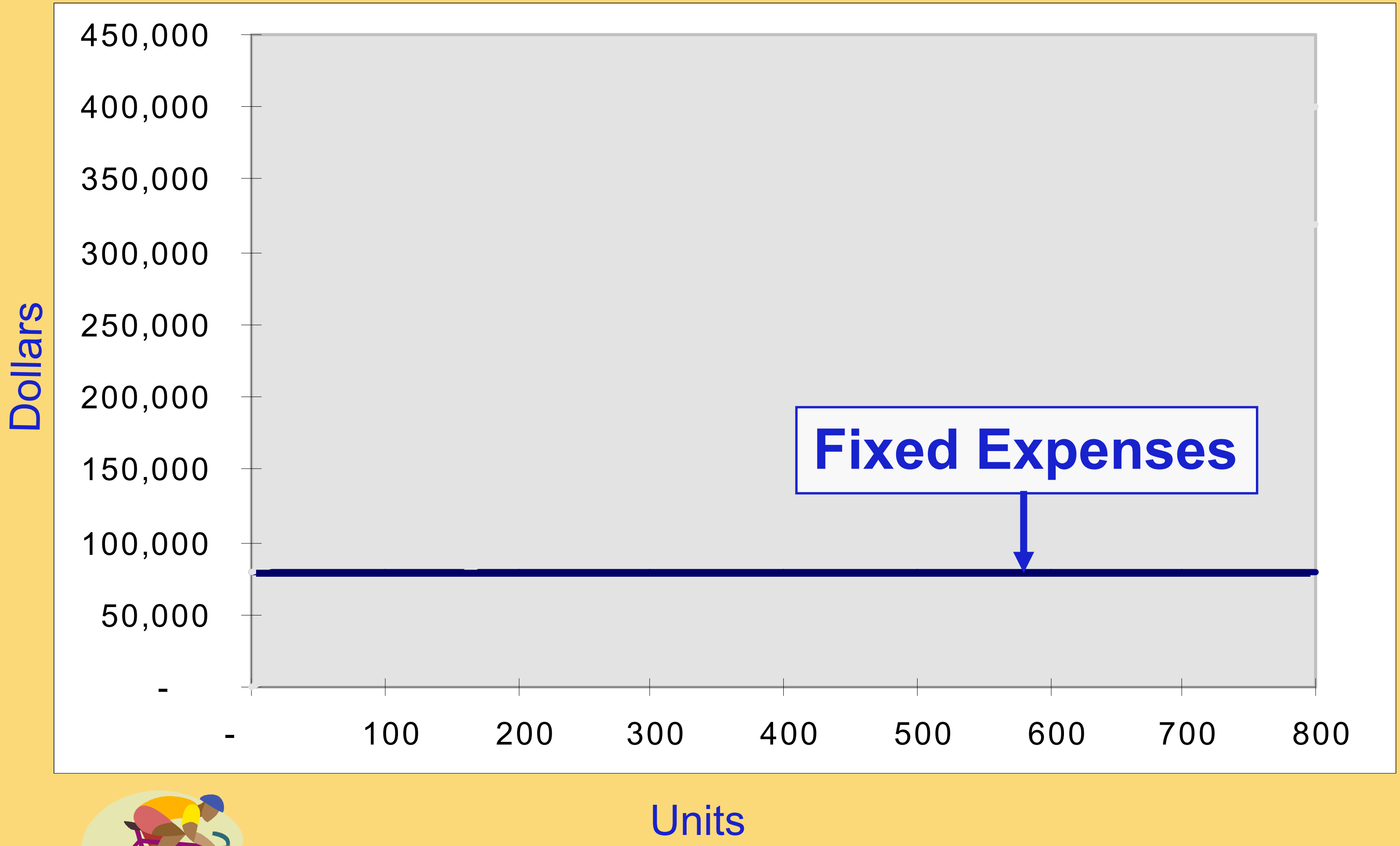
Prepare and interpret a cost-volume-profit (CVP) graph.

CVP Relationships in Graphic Form

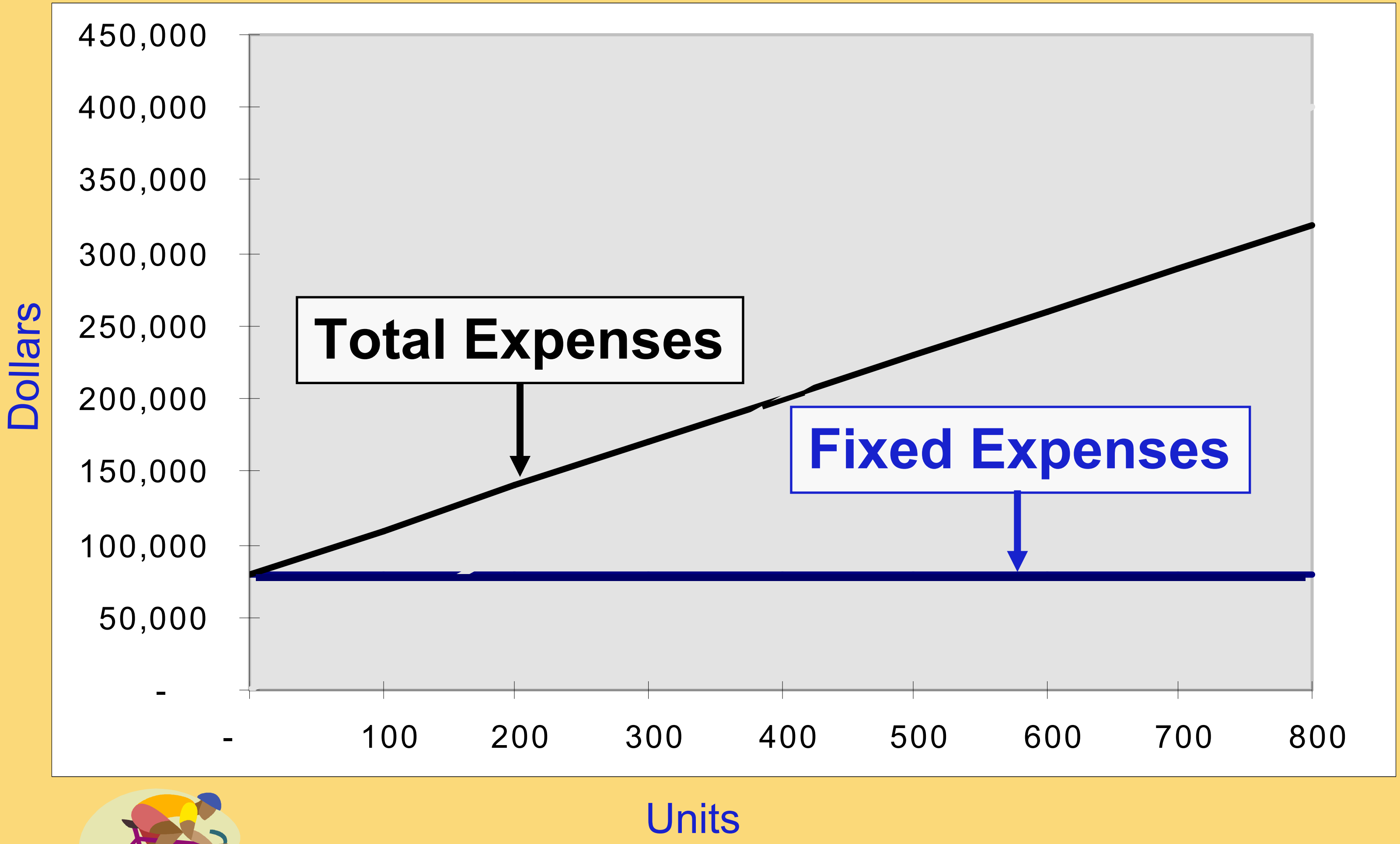
The relationship among revenue, cost, profit and volume can be expressed graphically by preparing a CVP graph. Racing developed contribution margin income statements at 300, 400, and 500 units sold. We will use this information to prepare the CVP graph.

	Income 300 units	Income 400 units	Income 500 units
Sales	$ 150,000	$ 200,000	$ 250,000
Less: variable expenses	90,000	120,000	150,000
Contribution margin	$ 60,000	$ 80,000	$ 100,000
Less: fixed expenses	80,000	80,000	80,000
Net operating income	$ (20,000)	$ -	$ 20,000

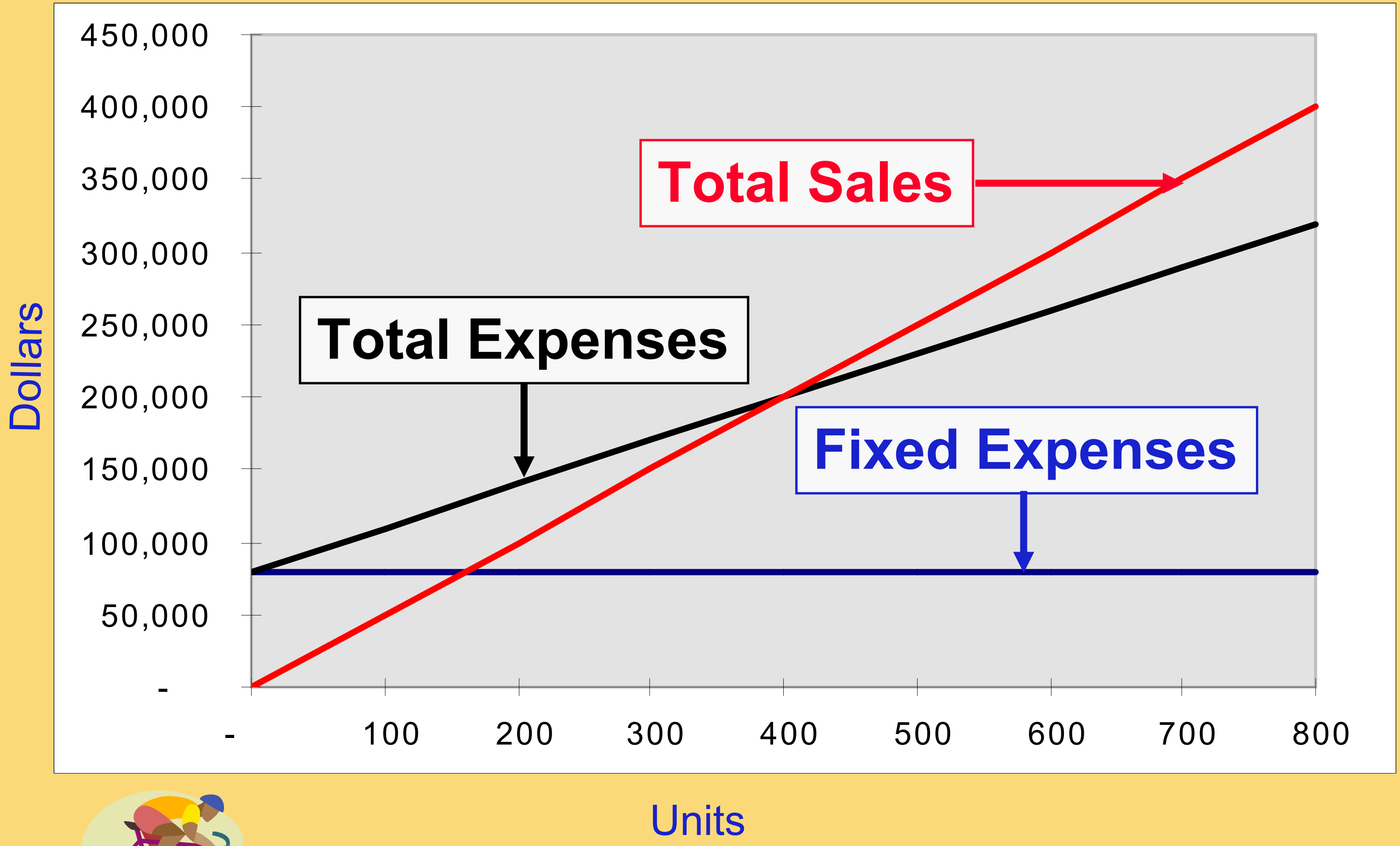
CVP Graph


450,000
400,000
350,000
300,000
250,000
200,000
150,000
100,000
50,000
-
Dollars
-
100
200
300
400
500
600
700
800
Fixed Expenses
Units

CVP Graph

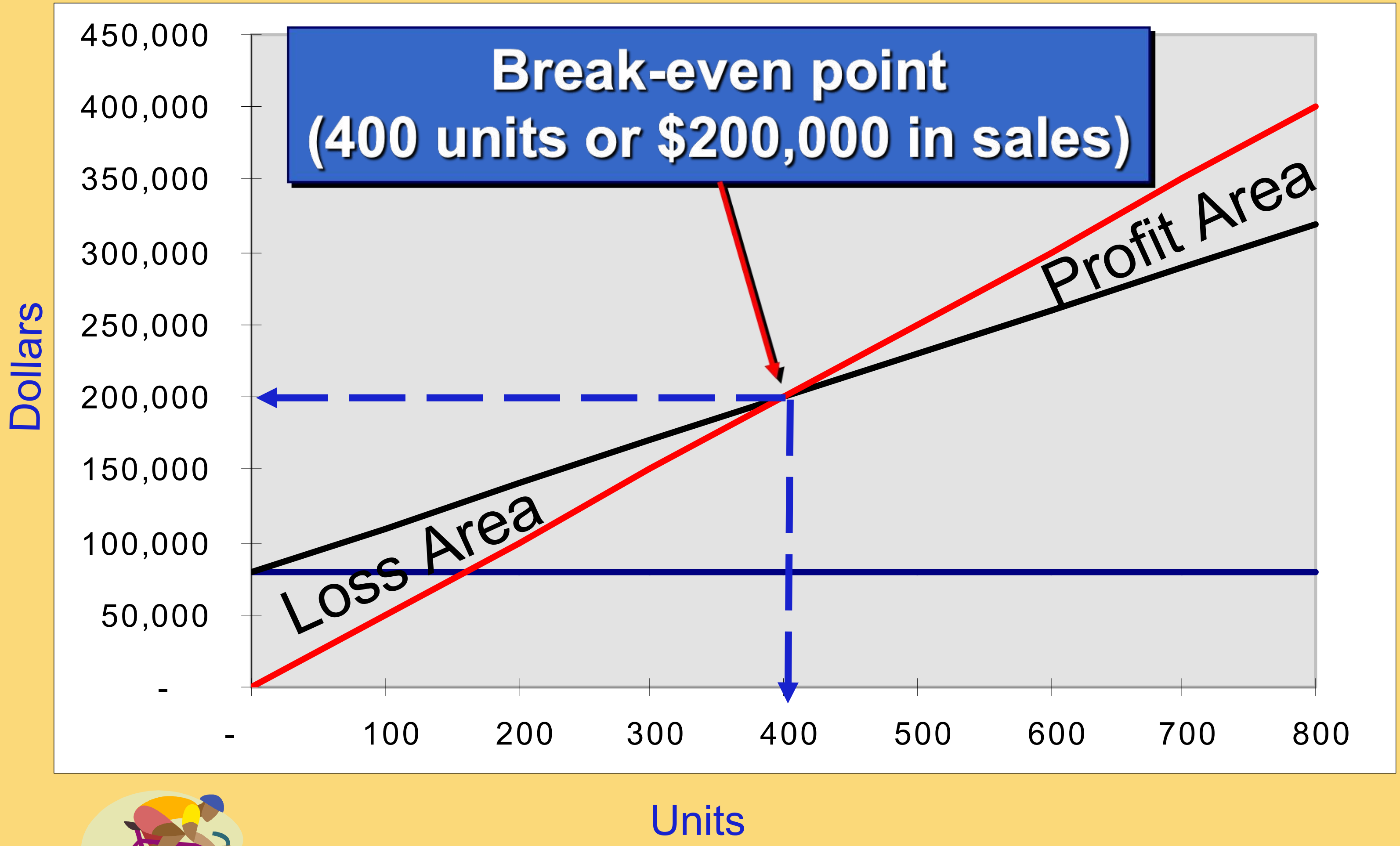

450,000
400,000
350,000
300,000
250,000
200,000
150,000
100,000
50,000
-
Dollars
Total Expenses
Fixed Expenses
-
100
200
300
400
500
600
700
800
Units

CVP Graph



450,000
400,000
350,000
300,000
250,000
200,000
150,000
100,000
50,000
-
Dollars
Total Sales
Total Expenses
Fixed Expenses
-
100
200
300
400
500
600
700
800
Units

CVP Graph


Break-even point
(400 units or $200,000 in sales)
450,000
400,000
350,000
300,000
250,000
200,000
150,000
100,000
50,000
-
Dollars
Profit Area
Loss Area
-
100
200
300
400
500
600
700
800
Units

Learning Objective 3

Use the contribution margin ratio (CM ratio) to compute changes in contribution margin and net operating income resulting from changes in sales volume.

Contribution Margin Ratio

The contribution margin **ratio** is:

$$\textbf{CM Ratio} = \frac{\textbf{Total CM}}{\textbf{Total sales}}$$

For Racing Bicycle Company the ratio is:

$$\frac{\$80,000}{\$200,000} = 40\%$$

Each $1.00 increase in sales results in a total contribution margin increase of 40¢.

Contribution Margin Ratio

Or, in terms of **units**, the contribution margin **ratio** is:

$$\text{CM Ratio} = \frac{\text{Unit CM}}{\text{Unit selling price}}$$

For Racing Bicycle Company the ratio is:

$$\frac{\$200}{\$500} = 40\%$$

Contribution Margin Ratio

	400 Bikes	500 Bikes
Sales	$ 200,000	$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net operating income	$ -	$ 20,000

A $50,000 increase in sales revenue results in a $20,000 increase in CM.
($50,000 × 40% = $20,000)

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is $1.49 and the average variable expense per cup is $0.36. The average fixed expense per month is $1,300. 2,100 cups are sold each month on average. What is the CM Ratio for Coffee Klatch?

a. 1.319

b. 0.758

c. 0.242

d. 4.139

Learning Objective 4

Show the effects on contribution margin of changes in variable costs, fixed costs, selling price, and volume.

Changes in Fixed Costs and Sales Volume

What is the profit impact if Racing can increase unit sales from 500 to 540 by increasing the monthly advertising budget by $10,000?

Changes in Fixed Costs and Sales Volume

$80,000 + $10,000 advertising = $90,000

Racing Bicycle Company
Contribution Income Statement
For the Month of June

	Current Sales (500 bikes)	Projected Sales (540 bikes)
Sales revenue	$ 250,000	$ 270,000
Less: Variables expenses	150,000	162,000
Contribution margin	100,000	108,000
Less: Fixed expenses	80,000	90,000
Net income	$ 20,000	$ 18,000

Sales *increased* by $20,000, but net operating income *decreased* by $2,000.

Changes in Fixed Costs and Sales Volume

The Shortcut Solution

Increase in CM (40 units X $200)	$ 8,000
Increase in advertising expenses	10,000
Decrease in net operating income	$ (2,000)

Change in Variable Costs and Sales Volume

What is the profit impact if Racing can use higher quality raw materials, thus increasing variable costs per unit by $10, to generate an increase in unit sales from 500 to 580?

Change in Variable Costs and Sales Volume

580 units × $310 variable cost/unit = $179,800

Racing Bicycle Company
Contribution Income Statement
For the Month of June

	Current Sales (500 bikes)	Projected Sales (580 bikes)
Sales revenue	$ 250,000	$ 290,000
Less: Variables expenses	150,000	179,800
Contribution margin	100,000	110,200
Less: Fixed expenses	80,000	80,000
Net income	$ 20,000	$ 30,200

Sales *increase* by $40,000, and net operating income *increases* by $10,200.

Change in Fixed Cost, Sales Price and Volume

What is the profit impact if Racing (1) cuts its selling price $20 per unit, (2) increases its advertising budget by $15,000 per month, and (3) increases sales from 500 to 650 units per month?

Change in Fixed Cost, Sales Price and Volume

Racing Bicycle Company
Contribution Income Statement
For the Month of June

	Current Sales (500 bikes)	Projected Sales (650 bikes)
Sales revenue	$ 250,000	$ 312,000
Less: Variables expenses	150,000	195,000
Contribution margin	100,000	117,000
Less: Fixed expenses	80,000	95,000
Net income	$ 20,000	$ 22,000

Sales *increase* by $62,000, fixed costs increase by $15,000, and net operating income *increases* by $2,000.

Change in Variable Cost, Fixed Cost and Sales Volume

What is the profit impact if Racing (1) pays a $15 sales commission per bike sold instead of paying salespersons flat salaries that currently total $6,000 per month, and (2) increases unit sales from 500 to 575 bikes?

Change in Variable Cost, Fixed Cost and Sales Volume

Racing Bicycle Company
Contribution Income Statement
For the Month of June

	Current Sales (500 bikes)	Projected Sales (575 bikes)
Sales revenue	$ 250,000	$ 287,500
Less: Variables expenses	150,000	181,125
Contribution margin	100,000	106,375
Less: Fixed expenses	80,000	74,000
Net income	$ 20,000	$ 32,375

Sales *increase* by $37,500, variable costs *increase* by $31,125, but fixed expenses *decrease* by $6,000.

Change in Regular Sales Price

If Racing has an opportunity to sell 150 bikes to a wholesaler without disturbing sales to other customers or fixed expenses, what price would it quote to the wholesaler if it wants to increase monthly profits by $3,000?

Change in Regular Sales Price

$ 3,000 ÷ 150 bikes	=	$ 20	per bike
Variable cost per bike	=	300	per bike
Selling price required	=	$ 320	per bike

150 bikes × $320 per bike	=	$ 48,000
Total variable costs	=	45,000
Increase in net income	=	$ 3,000

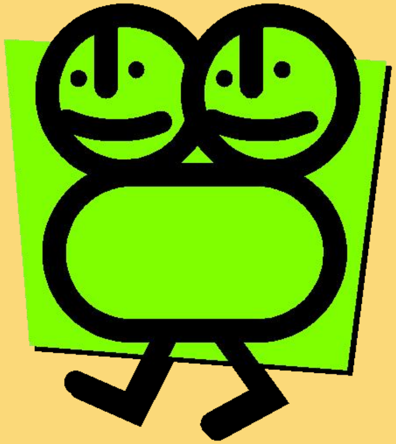
Learning Objective 5

Compute the break-even point in unit sales and sales dollars.

Break-Even Analysis

Break-even analysis can be approached in two ways:

1. Equation method
2. Contribution margin method



Equation Method

Profits = (Sales – Variable expenses) – Fixed expenses

OR

Sales = Variable expenses + Fixed expenses + Profits

At the break-even point profits equal zero

Break-Even Analysis

Here is the information from Racing Bicycle Company:

	Total	Per Unit	Percent
Sales (500 bikes)	$ 250,000	$ 500	100%
Less: variable expenses	150,000	300	60%
Contribution margin	$ 100,000	$ 200	40%
Less: fixed expenses	80,000		
Net operating income	$ 20,000		

Equation Method

We calculate the break-even point as follows:

Sales = Variable expenses + Fixed expenses + Profits

\$500Q = \$300Q + \$80,000 + \$0

Where:

Q = Number of bikes sold
\$500 = Unit selling price
\$300 = Unit variable expense
\$80,000 = Total fixed expense

Equation Method

We calculate the break-even point as follows:

Sales = Variable expenses + Fixed expenses + Profits

$$\$500Q = \$300Q + \$80{,}000 + \$0$$

$$\$200Q = \$80{,}000$$

$$Q = \$80{,}000 \div \$200 \text{ per bike}$$

$$Q = 400 \text{ bikes}$$

Equation Method

The equation can be modified to calculate the break-even point in sales dollars.

Sales = Variable expenses + Fixed expenses + Profits

$$X = 0.60X + \$80,000 + \$0$$

Where:

X = Total sales dollars

0.60 = Variable expenses as a % of sales

$80,000 = Total fixed expenses

Equation Method

The equation can be modified to calculate the break-even point in sales dollars.

Sales = Variable expenses + Fixed expenses + Profits

$$X = 0.60X + \$80{,}000 + \$0$$
$$0.40X = \$80{,}000$$
$$X = \$80{,}000 \div 0.40$$
$$X = \$200{,}000$$

Contribution Margin Method

The contribution margin method has two key equations.

$$\text{Break-even point in units sold} = \frac{\text{Fixed expenses}}{\text{CM per unit}}$$

$$\text{Break-even point in total sales dollars} = \frac{\text{Fixed expenses}}{\text{CM ratio}}$$

Contribution Margin Method

Let’s use the contribution margin method to calculate the break-even point in total sales dollars at Racing.

Break-even point in total sales dollars = $\dfrac{\text{Fixed expenses}}{\text{CM ratio}}$

$\dfrac{\$80{,}000}{40\%}$ = \$200,000 break-even sales

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is $1.49 and the average variable expense per cup is $0.36. The average fixed expense per month is $1,300. 2,100 cups are sold each month on average. What is the break-even sales in units?

a. 872 cups

b. 3,611 cups

c. 1,200 cups

d. 1,150 cups

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is $1.49 and the average variable expense per cup is $0.36. The average fixed expense per month is $1,300. 2,100 cups are sold each month on average. What is the break-even sales in dollars?

a. $1,300

b. $1,715

c. $1,788

d. $3,129

Learning Objective 6

Determine the level of sales needed to achieve a desired target profit.

Target Profit Analysis

The equation and contribution margin methods can be used to determine the sales volume needed to achieve a target profit.

Suppose Racing Bicycle Company wants to know how many bikes must be sold to earn a profit of $100,000.

The CVP Equation Method

Sales = Variable expenses + Fixed expenses + Profits

$$\$500Q = \$300Q + \$80{,}000 + \$100{,}000$$

$$\$200Q = \$180{,}000$$

$$Q = \textit{900 bikes}$$

The Contribution Margin Approach

The contribution margin method can be used to determine that 900 bikes must be sold to earn the target profit of $100,000.

$$\text{Unit sales to attain the target profit} = \frac{\text{Fixed expenses} + \text{Target profit}}{\text{CM per unit}}$$

$$\frac{\$80{,}000 + \$100{,}000}{\$200/\text{bike}} = \textit{900 bikes}$$

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is $1.49 and the average variable expense per cup is $0.36. The average fixed expense per month is $1,300. How many cups of coffee would have to be sold to attain target profits of $2,500 per month?

a. 3,363 cups

b. 2,212 cups

c. 1,150 cups

d. 4,200 cups

Learning Objective 7

Compute the margin of safety and explain its significance.

The Margin of Safety

The margin of safety is the excess of budgeted (or actual) sales over the break-even volume of sales.

Margin of safety = Total sales - Break-even sales

Let's look at Racing Bicycle Company and determine the margin of safety.

The Margin of Safety

If we assume that Racing Bicycle Company has actual sales of $250,000, given that we have already determined the break-even sales to be $200,000, the margin of safety is $50,000 as shown.

	Break-even sales 400 units	Actual sales 500 units
Sales	$ 200,000	$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net operating income	$ -	$ 20,000

The Margin of Safety

The margin of safety can be expressed in terms of the number of units sold. The margin of safety at Racing is \$50,000, and each bike sells for \$500.

$$\text{Margin of Safety in units} = \frac{\$50{,}000}{\$500} = 100 \text{ bikes}$$

The Margin of Safety

The margin of safety can be expressed as **20%** of sales.
($50,000 ÷ $250,000)

	Break-even sales 400 units	Actual sales 500 units
Sales	$ 200,000	$ 250,000
Less: variable expenses	120,000	150,000
Contribution margin	80,000	100,000
Less: fixed expenses	80,000	80,000
Net operating income	$ -	$ 20,000

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is $1.49 and the average variable expense per cup is $0.36. The average fixed expense per month is $1,300. 2,100 cups are sold each month on average. What is the margin of safety?

a. 3,250 cups

b. 950 cups

c. 1,150 cups

d. 2,100 cups

Cost Structure and Profit Stability

Cost structure refers to the relative proportion of fixed and variable costs in an organization. Managers often have some latitude in determining their organization's cost structure.

Cost Structure and Profit Stability

There are advantages and disadvantages to high fixed cost (or low variable cost) and low fixed cost (or high variable cost) structures.

An advantage of a high fixed cost structure is that income will be higher in good years compared to companies with lower proportion of fixed costs.

A disadvantage of a high fixed cost structure is that income will be lower in bad years compared to companies with lower proportion of fixed costs.

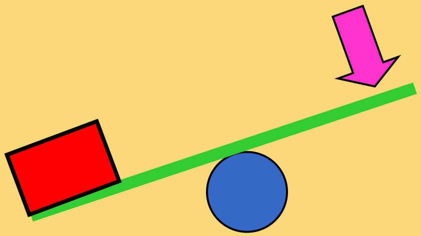
Learning Objective 8

Compute the degree of operating leverage at a particular level of sales and explain how it can be used to predict changes in net operating income.

Operating Leverage

A measure of how sensitive net operating income is to percentage changes in sales.

$$\text{Degree of operating leverage} = \frac{\text{Contribution margin}}{\text{Net operating income}}$$



Operating Leverage

At Racing, the degree of operating leverage is 5.

	Actual sales 500 Bikes
Sales	$ 250,000
Less: variable expenses	150,000
Contribution margin	100,000
Less: fixed expenses	80,000
Net income	$ 20,000

$$\frac{\$100{,}000}{\$20{,}000} = 5$$

Operating Leverage

With an operating leverage of 5, if Racing increases its sales by 10%, net operating income would increase by 50%.

Percent increase in sales		10%
Degree of operating leverage	×	5
Percent increase in profits		50%

Here's the verification!

Operating Leverage

	Actual sales (500)	Increased sales (550)
Sales	$ 250,000	$ 275,000
Less variable expenses	150,000	165,000
Contribution margin	100,000	110,000
Less fixed expenses	80,000	80,000
Net operating income	$ 20,000	$ 30,000

10% increase in sales from $250,000 to $275,000 . . .

. . . results in a 50% increase in income from $20,000 to $30,000.

Quick Check ✓

Coffee Klatch is an espresso stand in a downtown office building. The average selling price of a cup of coffee is $1.49 and the average variable expense per cup is $0.36. The average fixed expense per month is $1,300. 2,100 cups are sold each month on average. What is the operating leverage?

a. 2.21

b. 0.45

c. 0.34

d. 2.92

Quick Check ✓

At Coffee Klatch the average selling price of a cup of coffee is $1.49, the average variable expense per cup is $0.36, the average fixed expense per month is $1,300 and an average of 2,100 cups are sold each month.

If sales increase by 20%, **by** how much should net operating income increase?

a. 30.0%

b. 20.0%

c. 22.1%

d. 44.2%

Learning Objective 9

Compute the break-even point for a multiproduct company and explain the effects of shifts in the sales mix on contribution margin and the break-even point.

The Concept of Sales Mix

- Sales mix is the relative proportion in which a company's products are sold.
- Different products have different selling prices, cost structures, and contribution margins.

Let's assume Racing Bicycle Company sells bikes and carts and that the sales mix between the two products remains the same.

Multi-product break-even analysis

Racing Bicycle Co. provides the following information:

	Bicycles		Carts		Total	
Sales	$ 250,000	100%	$ 300,000	100%	$ 550,000	100.0%
Variable expenses	150,000	60%	135,000	45%	285,000	51.8%
Contribution margin	$ 100,000	40%	$ 165,000	55%	265,000	48.2%
Fixed expenses					170,000	
Net operating income					$ 95,000	
Sales mix	$ 250,000	45%	$ 300,000	55%	$ 550,000	100%

$$\frac{\$265{,}000}{\$550{,}000} = 48.2\% \text{ (rounded)}$$

Multi-product break-even analysis

$$\text{Break-even sales} = \frac{\text{Fixed expenses}}{\text{CM Ratio}}$$

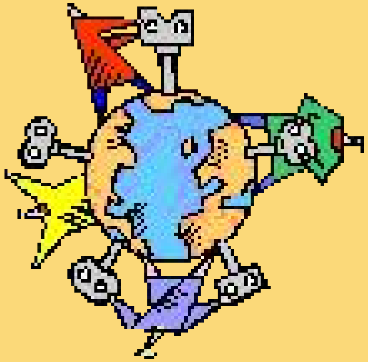
$$= \frac{\$170{,}000}{48.2\%}$$

$$= \$352{,}697$$

	Bicycles		Carts		Total	
Sales	$ 158,714	100%	$ 193,983	100%	$ 352,697	100.0%
Variable expenses	95,228	60%	87,292	45%	182,521	51.8%
Contribution margin	$ 63,486	40%	$ 106,691	55%	170,176	48.2%
Fixed expenses					170,000	
Net operating income			Rounding error →		$ 176	
Sales mix	$ 158,714	45%	$ 193,983	55%	$ 352,697	100%

Key Assumptions of CVP Analysis

1. Selling price is constant.
2. Costs are linear.
3. In multiproduct companies, the sales mix is constant.
4. In manufacturing companies, inventories do not change (units produced = units sold).



End of Chapter 6